

KUNAL PANDEY

Software Developer Co-op | AI & Cloud

Email: kunal.pandey.work@outlook.com | Phone: (647) 451-0445

LinkedIn: www.linkedin.com/in/kunal-pandey25 | Portfolio: portfolio-pandey-kunal.vercel.app/

Professional Summary

Software developer and AI/cloud student with hands-on experience building backend APIs, GenAI applications, cloud-deployed systems, and user-facing tools. Skilled in Python, JavaScript, TypeScript, FastAPI, Flask, REST APIs, Docker, AWS, GCP, GitHub Actions CI/CD, testing, debugging, and technical documentation. Interested in SDLC, application integration, cloud computing, microservices, and reliable software delivery.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, HTML5/CSS3

Backend & APIs: FastAPI, Flask, Streamlit, REST APIs, JSON, API Integration, Microservices Fundamentals

Frontend: React, Next.js, Tailwind CSS, Responsive UI, Component-Based Development

Cloud & DevOps: AWS EC2/ECR/ECS, GCP Cloud Run, Docker, GitHub Actions CI/CD

Software Development: SDLC, Debugging, Troubleshooting, Technical Documentation, Release Management Fundamentals

Testing & Tools: Pytest, Unit Testing, Test-Driven Development Fundamentals, Git, GitHub

AI & Data: LangChain, RAG, Pinecone, Redis, Prompt Engineering, Hugging Face, Gemini/Vertex AI, Power BI, Excel

ML: Scikit-learn, PyTorch fundamentals, Pandas, NumPy, Feature Engineering, Model Evaluation, MLflow

PROJECTS

Domain RAG Assistant | Python, LangChain, Redis, Pinecone, Docker, AWS EC2 | [GitHub](#)

- Built a multi-domain RAG assistant for Medical, ML, and LLM knowledge bases using Hugging Face embeddings, Pinecone, and metadata-based domain filtering; achieved 12/12 passing retrieval evaluation tests across domain accuracy, source citation, and wrong-domain blocking.
- Implemented Redis-backed conversation memory via LangChain RunnableWithMessageHistory, enabling coherent multi-turn sessions across browser restarts with configurable TTL.
- Built and tested AI workflow components for retrieval, memory, domain filtering, response grounding, and wrong-domain detection to improve reliability across multi-domain interactions.
- Containerized the full stack (Flask + Pinecone + Redis + Gemini) with Docker and deployed to AWS EC2 via GitHub Actions CI/CD automating build, ECR push, and container relaunch on every push to main.

AI Job Recommendation System | Python, LLMs, Streamlit, Resume Parsing, APIs | [GitHub](#)

- Engineered and deployed a full-stack GenAI career assistant on Google Cloud Run parsing resume PDFs with PyMuPDF, orchestrating analysis via Gemini on Vertex AI, and delivering 4 structured LLM outputs (resume snapshot, skill-gap report, 30-day roadmap, job matches) end-to-end in under 30 seconds.
- Built a real-time job fetching and recommendation pipeline integrating the Apify Global / All-Source Jobs API, computing match scores, matched skills, missing skills, and natural-language recommendation explanations across 10+ job categories.

- Implemented production-grade deployment: Docker containerization, Google Secret Manager for API credential management, GitHub Actions CI/CD for automated redeployment on every push, and Vertex AI IAM service account configuration.
- Shipped user-facing features including paginated job results, CSV export of recommendations, and a structured Streamlit interface live and publicly accessible via Google Cloud Run.

Housing Price Prediction System | Python, XGBoost, FastAPI, Docker, AWS | [GitHub](#)

- Trained an XGBoost regression model on 800K+ rows of U.S. housing time-series data, achieving R^2 of 0.87 after Optuna hyperparameter optimization across 50+ trial configurations.
- Developed reusable Python workflows for data preparation, feature engineering, model training, inference, and MLflow experiment tracking to support the end-to-end ML lifecycle.
- Evaluated model performance across 10+ experiments using RMSE, MAE, and R^2 , documenting results to compare model behavior and support model selection decisions.
- Deployed the application using Docker, FastAPI, AWS ECS/ECR, and GitHub Actions CI/CD, serving predictions through a REST API with an interactive Streamlit visualization front-end.

PROFESSIONAL EXPERIENCE

Overnight Associate | Walmart | April 2026 – Present

- Maintained high data accuracy in inventory tracking and stock monitoring while managing large product volumes under tight time constraints.
- Demonstrated strong attention to detail and independent work ethic, consistently meeting nightly productivity targets while pursuing full-time technical project development.
- Collaborated as part of a cohesive store team to maintain overall visual and operational store standards, adapting flexibly to changing priorities and business needs.

Overnight Associate | Metro | Nov 2024 – May 2025

- Replenished shelves to ensure products were available and easy for customers to find.
- Applied FIFO inventory management and stock monitoring across high-SKU sections, ensuring accurate product placement and availability.
- Worked independently with minimal supervision while maintaining operational accuracy and meeting nightly throughput goals.

EDUCATION

Postgraduate Certificate – Artificial Intelligence | Seneca Polytechnic, North York, Ontario

Postgraduate Diploma – Cloud Computing | Cambrian College, Brampton, Ontario

GPA: 4.0 / 4.0

Bachelor of Technology – Electronics and Communication | Manipal University, India

GPA: 3.4 / 4.0